Encoder odometry heading scale in 90° junction turns

Measured error, repeatability, and implications for the digital twin

SUBSYSTEM AND INVESTIGATION AIM

Odometry is the only heading source available to our maze line-following robot between junctions. The local model converts wheel counts to heading using a wheel radius $r = 16$ mm, wheel separation $b = 90$ mm and $N = 358.3$ counts/rev (Pololu 3Pi+); the digital twin uses the same relationship. At every 90° junction the robot therefore relies on the encoder-to-heading scale; an error in that scale accumulates along the route and makes the twin's pose prediction diverge from the real robot.

This report asks how closely the encoder heading matches the physical heading during a 90° in-place turn, whether the error is repeatable, and what effective wheel separation explains it. If the nominal geometry were correct, the encoder heading θ_{odom} would be 90° with zero mean error; a systematic offset would instead identify a geometry calibration error.

MEASUREMENT APPROACH

Firmware in `Controller.cpp` waits for 500 ms with no encoder change, resets odometry, then drives both wheels at ± 35 PWM until the differential count reaches 503.86 counts, the nominal 90° target $\Delta N = \theta b N/(2\pi r)$. It then stops for 1.5 s. Ten trials (five left, five right, alternating) were run on one A4 sheet taped to a desk, with the robot centred on a printed cross and the same cable routing. Trial 5 was rejected after the USB cable snagged; the 4 s safety timeout stopped it at 16.5°.

The independent reference is the onboard LSM6DS33 gyro (104 Hz, ± 245 dps). Its z rate is integrated and the bias is re-estimated during each 500 ms quiet window. Per trial we compute θ_{odom} at the stop and after settling, θ_{true} from the gyro, the direction-corrected error $e = \text{sgn}(\theta)(\theta_{\text{odom}} - \theta_{\text{true}})$, and $b_{\text{eff}} = \Delta N\,(2\pi r/N)/|\theta_{\text{true}}|$. The gyro scale was checked against a printed protractor (agreement within about 3°); the remaining uncertainty is slip and coast.

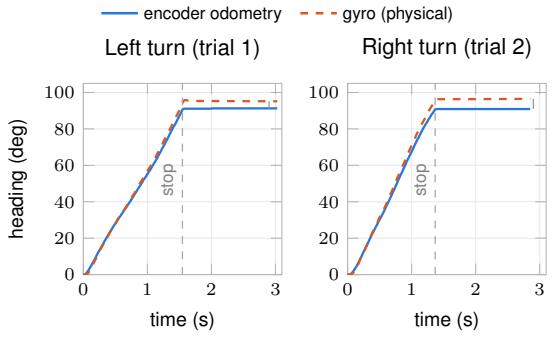

Figure 1. Encoder and gyro headings for one left (trial 1) and one right (trial 2) turn. The dashed line is the stop; afterwards the gyro coasts while the encoder changes only with the remaining counts.

RESULTS AND INTERPRETATION

Figure 1 shows representative left and right trials. The two headings track during the turn but separate by the stop: in trial 1 the encoder reports 90.6° while the gyro has reached 94.2°. The gyro then continues to change after the stop, whereas the encoder only accumulates the remaining wheel counts.

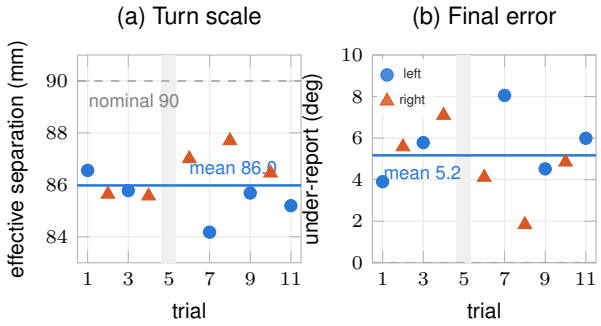

Figure 2. (a) Effective separation from counts and gyro at the stop. (b) Final encoder under-report after settling. Circles: left; triangles: right; shaded band: rejected trial 5; solid/dashed: mean and nominal/zero.

Across the ten valid trials, θ_{odom} at the stop was 90.52 ± 0.25 and the gyro 94.77 ± 1.1. The encoder under-reported the turn in every trial: 4.25 ± 1.09 at the stop (95% CI 3.47–5.03°) and 5.17 ± 1.75 after settling (CI 3.92–6.42°). The implied separation was 85.98 ± 0.99 mm (CI 85.27–86.69 mm), 4.47% below the nominal 90 mm (Figure 2a). Rescaling the odometry by $90/85.98$ removes the stop bias (mean -0.01, SD 1.09); the final residual is -0.92 ± 1.76 because the coast rotation is not counted. No significant left/right difference was found ($p = 0.135$ at the stop, $p = 0.437$ after settling): the scale error is a direction-independent geometry error with about 1° (1.2%) random variation.

IMPLICATIONS FOR THE DIGITAL TWIN

The twin instantiates the same model with $r = 16$ mm and $b = 90$ mm (`DigitalTwin/Simulator.pde`), using $\Delta\theta = (d_R - d_L)/b$. Setting $b = 86.0$ mm, or scaling θ by 1.047, removes the 4.5% bias at the stop. With $b = 90$ mm, four successive 90° turns accumulate about 17° of heading error at the stop (about 21° after coast); the simulated robot would predict a different corridor from the real one. The twin should also represent the short post-stop coast (mean 1.3°, up to 3.3°); integrating counts alone under-predicts the final heading by about 1° per turn. A check is five fresh 90° turns with $b = 86.0$ mm; a mean residual above 1° or an SD above 2° would require revisiting the calibration or coast model. The result applies to in-place turns at PWM 35 on paper; other conditions need a separate check.

Evidence and sources. Code: `MyFirstRobot/Controller.*`, `MyFirstRobot.ino`, `OdometryModel.*`. Data: `test_materials/turn_capture_main.txt` (trial 5 rejected); `results/turn_trial_summary.csv`. Analysis: `analysis/analyse_turn_trials.py`.